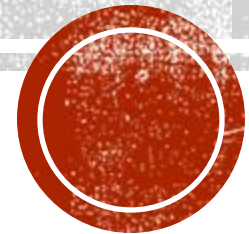


CLOUD COMPUTING

Chandara Chea

chandara.chea@cadt.edu.kh



WEEK 3 — STORAGE SOLUTIONS IN CLOUD

- **Learning Objectives:**

- Understand what is cloud storage
- Understand difference between ephemeral and persistent storage
- Understand most common persistent storage type
- Introduce AWS S3



WHAT IS CLOUD STORAGE?

- **Cloud storage** refers to a model of data storage in which data is stored on remote servers that are hosted and managed by cloud service providers.
- These servers are accessible over the internet, meaning you can store, access, and manage your data remotely without the need to own or maintain physical storage hardware.
- Cloud storage is scalable, flexible, and can be used for a variety of purposes, including backup, archiving, file sharing, and running applications.
- There are 2 important distinctions in cloud storage: ephemeral and persistent.



EPHEMERAL STORAGE

- Ephemeral storage is temporary, and data is lost when the resource (such as a virtual machine or container) is stopped or terminated. It's often used for temporary data that does not need to persist beyond the life of the instance or session.
- Characteristics:
 - Short-lived: The data exists only as long as the associated resource (e.g., virtual machine or container) is running.
 - No Data Persistence: If the resource is restarted or shut down, the data is lost.
 - Low-Cost: Ephemeral storage is typically less expensive than persistent storage because it does not offer the same durability and availability guarantees.
- Use Cases:
 - Caching: Storing temporary data that speeds up operations (e.g., web server caches).
 - Temporary Files: For intermediate computation results or temporary processing that doesn't need to be saved long-term.
 - Containers: Containers, like Docker, often use ephemeral storage because the data is not typically saved once the container stops.
- Examples: Amazon EC2 Instance Store - AWS provides ephemeral storage with EC2 instances. Data stored on an instance store is lost when the instance is terminated.



PERSISTENT STORAGE

- Persistent storage refers to storage that retains data even after the associated resource (like a virtual machine or container) is stopped or terminated. This type of storage is designed for long-term data retention and is more reliable for data that must not be lost.
- Characteristics:
 - Long-lived: Data remains intact even if the associated compute resource is stopped or restarted.
 - Durability: It is typically designed for high durability, ensuring that data remains available even in the event of hardware failure or outages.
 - Cost: Persistent storage is more expensive than ephemeral storage due to its durability and availability guarantees.
- Use Cases:
 - Database Storage: For critical data that must be saved between application runs (e.g., databases, transactional systems).
 - User Files: Storing user data that must persist across sessions, like profiles, documents, and settings.
 - Backup: For long-term backups or archives that need to survive system failures or shutdowns.
 - Enterprise Applications: For applications that require the data to persist and be recoverable after failures.



BLOCK STORAGE

- Persistent storage offers different types of storage for different use cases, such as file storage, block storage, object storage, and more.
- Block storage is a low-level data storage mechanism that organizes data into fixed-size blocks, much like how data is managed on traditional hard drives (HDDs) and solid-state drives (SSDs).
- Block storage behaves like a raw disk, meaning that the operating system (OS) or application is responsible for managing how data is organized within the blocks (e.g., file systems like NTFS, ext4).
- It's ideal for applications that require low-latency access to data.
- Use Cases: Virtual machine disks, database storage, high-performance applications.
- Examples: Amazon EBS (Elastic Block Store)- AWS block storage solution for EC2 instances, providing persistent block-level storage.



FILE STORAGE

- File storage uses a hierarchical file system (similar to traditional operating systems) to store and organize data. Data is stored in directories (folders) and accessed using file paths. File storage is commonly used for applications that need to access data as files, such as content management systems or shared file systems.
- File storage allows for file-based access using protocols such as NFS (Network File System). These protocols enable systems to read/write files and manage file systems across multiple systems or networks.
- Many file storage solutions allow multiple users or systems to access the same file system concurrently, making it useful for collaborative environments.
- Example: Amazon EFS (Elastic File System)- AWS's managed file storage service that can be mounted across multiple EC2 instances.



OBJECT STORAGE

- Object storage is a highly scalable and flexible storage model that stores data as objects rather than blocks or files. Each object consists of three components: the data itself, metadata, and a unique identifier (object ID). This makes object storage ideal for storing large amounts of unstructured data, such as images, videos, and backups.
- Each object has metadata that provides descriptive information about the object, such as its creation date, owner, and type of data. This allows for better indexing and searching of objects.
- Object storage is accessed using a simple API (usually HTTP-based, like RESTful APIs), making it well-suited for cloud applications and content distribution.
- Example: Amazon S3 (Simple Storage Service) - a scalable object storage service that stores unstructured data, such as backups, videos, and logs.



USE CASE SCENARIOS

- Scenario 1:
A website runs on 5 EC2 instances. All servers must access the same images and HTML files.
- Scenario 2:
A university wants to store thousands of assignment PDFs for future reference.
- Scenario 3:
A database (like MySQL) on EC2 needs fast storage to store its data files.
- Scenario 4:
A mobile app lets users upload profile pictures and videos.



INTRODUCTION TO AWS S3

- https://www.youtube.com/watch?v=ecv-19sYL3w&ab_channel=AmazonWebServices
- **Amazon S3** is a **scalable object storage service** provided by AWS. It allows you to **store and retrieve any amount of data**, from anywhere on the web, at any time.
- **How S3 Works** (in Simple Terms)
 - **Buckets:**
You store your data in **buckets**. Think of a **bucket** as a **folder** in the cloud.
Example: **my-website-bucket** for all website data.
 - **Objects:**
Each file you upload is an **object**. Each object has a unique **key** and metadata.
Example: **profile-pic.jpg** or **video.mp4**.



AWS S3 STORAGE CLASSES

- **Standard:** General-purpose, high-performance storage for frequently accessed data.
- **Intelligent-Tiering:** Automatically moves data between two access tiers based on changing access patterns.
- **One Zone-IA:** For infrequently accessed data that doesn't need to be stored in multiple Availability Zones.
- **Glacier:** Low-cost storage for archival data. Retrieval takes time but is very cost-effective for long-term storage.



CONCLUSION

- Cloud storage has become an essential part of modern computing, enabling individuals and organizations to store, access, and manage data flexibly and efficiently.
- Ephemeral storage is a temporary data storage.
- Persistent storage consists of block storage, file storage and object storage.

